

Locking Away Prosperity? Evaluating the Economic Impact of the Key to NYC Program

1 Introduction

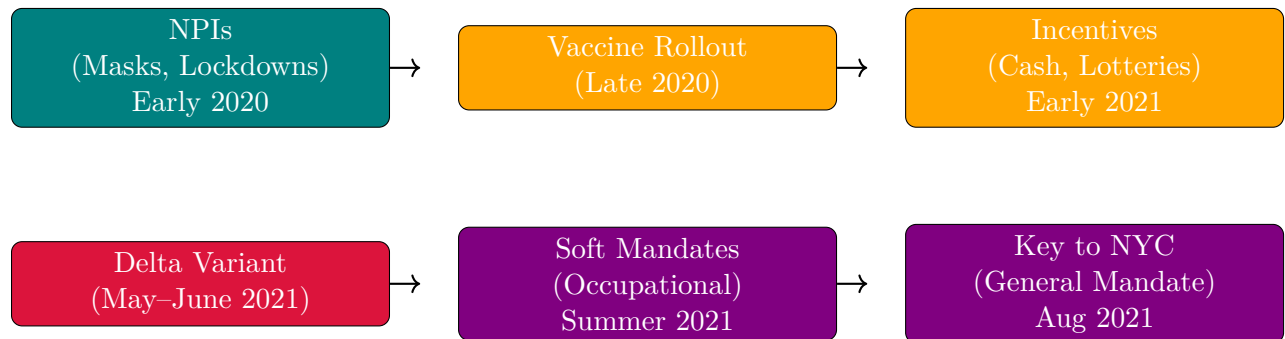
The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, [Friedson et al. \[2021\]](#) and [Yang \[2021\]](#) use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, [Bayat et al. \[2020\]](#) estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like [Gibson and Sun \[2023\]](#) and [Lee and Yang \[2022\]](#) document labor market costs of lockdowns, while [Ke and Hsiao \[2022\]](#) show that although strict NPIs incur short-term economic losses, those effects may be temporary. Vaccine mandates emerged as a less restrictive (but still controversial) policy tool during later phases of the pandemic. Unlike lockdowns, mandates aimed to curb transmission without shutting down businesses. Nevertheless, critics argued that mandates could hamper fragile economic recoveries by alienating customers or exacerbating staffing shortages in consumer-facing sectors like restaurants. Supporters countered that mandates would improve public safety and restore consumer confidence. These competing narratives came to a head with New York City’s *Key to NYC* program—implemented in August 2021—the first large-scale vaccine mandate for indoor commercial activity in the U.S. [[Towey, 2021](#), [Rogers, 2021](#), [Bardosh et al., 2022](#)].

This paper evaluates the causal effect of the Key to NYC mandate on employment in New York City’s full-service restaurant sector—the segment most directly affected by the policy. While prior research has analyzed mandates’ effects on vaccination rates and health outcomes [[Cohn et al., 2022](#), [Acharya and Dhakal, 2021](#), [Lang et al., 2023](#)], relatively little is known about their labor market consequences, particularly in industries that rely heavily on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, this paper adopts a synthetic control approach to estimate the counterfactual trajectory of employment absent the mandate. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct a synthetic control for NYC’s year-over-year

growth in full-service restaurant employment. The donor pool consists of 29 MSAs that did not implement similar mandates during the same period.¹

The rest of the paper is organized as follows. Section 2 provides background on the evolution of pandemic interventions and the Key to NYC program. Section 3 describes the employment data and outcome construction. Section 4 outlines the empirical strategy, including the synthetic control methodology. Section 5 presents the main findings and robustness checks. Section 6 interprets the results and discusses limitations. Section 7 concludes.

2 Policy Background



Timeline of major pandemic policy phases

2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The first and most famous of these interventions was the 2020 lockdown in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so, with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban. Mask mandates were widely adopted as well due to their low cost and direct mechanism of action [Hansen and Mano, 2023, Chernozhukov et al., 2021, Mitze et al., 2020]. While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they provoked political and economic backlash due to their disruptive nature.

2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older

¹For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

adults and (often mandated for) healthcare workers. However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination [Barber and West, 2022, Robertson et al., 2021]. These included direct cash payments, vaccine lotteries (e.g., Ohio’s “Vax-a-Million” from May to June [Fuller et al., 2022]), and smaller symbolic giveaways. One paper dubbed this the era of “donuts, drugs, booze, and guns,” reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements [Tinari and Riva, 2021].

Despite their creativity, most incentives had modest or short-lived effects [Saban et al., 2021, Walkey et al., 2021, Khazanov et al., 2023]. Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment [American Medical Association et al., 2021]. Over the summer, local governments and organizations expanded this approach [American Medical Association et al., 2021, Al Jazeera, 2021]. By August 2021, New York City introduced the Key to NYC program—one of the first large-scale vaccine mandates in the U.S. applied to the general public. Unlike prior efforts that relied on encouragement, Key to NYC made access to indoor dining, fitness, and entertainment conditional on vaccination status. It represented a new policy mechanism: exclusion rather than incentive.

2.3 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher. However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED). The sample runs from January 1991 through August 2022, yielding $T = 380$ monthly observations per unit. Units are indexed by $j \in \mathbb{N}$ and time by $t \in \mathbb{N}$. Unit $j = 1$ is the treated unit (New York City MSA). The donor pool is $\mathcal{N}_0 = \mathcal{N} \setminus \{1\}$ with cardinality N_0 ; in the baseline sample $N_0 = 29$ (so $N = 30$ total MSAs).

Let e_{jt} denote employment in full-service restaurants in unit j at month t . The primary outcome is the year-over-year growth rate (in percentage points),

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \quad j = 1, \dots, N, \quad t = 13, \dots, T.$$

Collect outcomes into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top \in \mathbb{R}^T$ and the donor matrix

$$\mathbf{Y}_0 = [\mathbf{y}_j]_{j \in \mathcal{N}_0} \in \mathbb{R}^{T \times N_0}.$$

The *Key to NYC* policy was implemented in August 2021. Let T_0 denote the last pre-treatment month (July 2021). Define the pre- and post-treatment index sets $\mathcal{T}_1 = \{t \leq T_0\}$ and $\mathcal{T}_2 = \{t > T_0\}$. For convenience we partition vectors and matrices into pre/post blocks, e.g. $\mathbf{y}_j = (\mathbf{y}_{j,\mathcal{T}_1}^\top, \mathbf{y}_{j,\mathcal{T}_2}^\top)^\top$.

Let $y_{jt}(0)$ and $y_{jt}(1)$ denote the potential outcomes for unit j at time t under no treatment and treatment, respectively. Define the observed treatment indicator

$$d_{jt} = \mathbf{1}\{j = 1 \wedge t > T_0\},$$

so only the treated unit is exposed in the post period. The observed outcome satisfies the switching equation

$$y_{jt} = (1 - d_{jt}) y_{jt}(0) + d_{jt} y_{jt}(1).$$

Consequently only one of $\{y_{jt}(0), y_{jt}(1)\}$ is observed for each (j, t) , and the causal effect for unit j at time t is

$$\tau_{jt} = y_{jt}(1) - y_{jt}(0).$$

Our primary object of interest is the post-treatment average treatment effect on the treated (ATT) for the treated unit $j = 1$:

$$\text{ATT} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} [y_{1t}(1) - y_{1t}(0)].$$

We additionally aggregate MSA growth rates to the level of Combined Statistical Areas (CSAs) in order to reduce noise, exploit the structure of contiguous labor markets, and construct more stable counterfactuals. Each donor MSA $j \in \mathcal{N}_0$ belongs to a CSA $c \in \mathcal{C}$. Let w_{jt} denote the relative employment weight of MSA j within its CSA at month t ,

$$w_{jt} = \frac{e_{j,t-12}}{\sum_{k \in \mathcal{N}_c} e_{k,t-12}}, \quad j \in \mathcal{N}_c,$$

where $\mathcal{N}_c \subseteq \mathcal{N}_0$ indexes MSAs in CSA c , and $e_{j,t-12}$ is the employment level lagged by 12 months. These weights satisfy $w_{jt} \geq 0$ and $\sum_{j \in \mathcal{N}_c} w_{jt} = 1$ for each t .

The Törnqvist (logarithmic) growth rate of CSA c at month t is then computed as the convex combination of its component MSAs' growth rates:

$$y_{ct}^{\text{CSA}} = \sum_{j \in \mathcal{N}_c} \frac{w_{jt} + w_{j,t-1}}{2} \cdot y_{jt}.$$

This approach allows the CSA growth rate to reflect the contributions of individual MSAs in a way that is proportional to their size while remaining insensitive to extreme employment levels, providing a theoretically grounded way to aggregate growth rates.

Stacking these vectors over $t = 13, \dots, T$ gives the CSA growth matrix

$$\mathbf{Y}_{\text{CSA}} = [\mathbf{y}_c^{\text{CSA}}]_{c \in \mathcal{C}} \in \mathbb{R}^{T \times |\mathcal{C}|}.$$

We can also express this using matrix notation. Define $\mathbf{W}_t \in \mathbb{R}^{N_0 \times |\mathcal{C}|}$ as the MSA-to-CSA membership/weight matrix at month t , with entries

$$(\mathbf{W}_t)_{jc} = \begin{cases} \frac{e_{j,t-12}}{\sum_{k \in \mathcal{N}_c} e_{k,t-12}} & j \in \mathcal{N}_c, \\ 0 & \text{otherwise.} \end{cases}$$

The CSA growth vector at month t is

$$\mathbf{y}_t^{\text{CSA}} = \mathbf{Y}_t^\top \bar{\mathbf{W}}_t,$$

where $\mathbf{Y}_t = (y_{1t}, \dots, y_{N_0 t})^\top$ and $\bar{\mathbf{W}}_t = (\mathbf{W}_t + \mathbf{W}_{t-1})/2$ is the matrix of averaged weights. Stacking these vectors over all t yields $\mathbf{Y}_{\text{CSA}}$, which provides CSA-level growth series suitable for constructing more stable synthetic controls.

4 Econometric Methodology

5 Results

6 Discussion

7 Conclusion

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